Yao Du

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Human-AI Interaction | Human Feedback Interfaces | Human Factors



EDUCATION

Southeast University (SEU) **Nanjing, China**
M.Eng. in Design, School of Mechanical Engineering Sept. 2023 – Jun. 2026
- Average Score: 89.4/100 | Focus: Human-AI interaction, ergonomics, experimental design

Chongqing University (CQU) **Chongqing, China**
B.S. in Industrial Design, College of Mechanical and Vehicle Engineering Sept. 2019 – Jun. 2023
- GPA: 3.57/4 (86.64/100) | Focus: interaction design, design psychology, product innovation

PUBLICATIONS

- Du, Y., Chen, Y., & Xue, C. (2026). Shared mental models as cognitive scaffolds: Optimizing interface design for human-in-the-loop alignment tasks. International Journal of Industrial Ergonomics, 114, 103955. **[2025 JCR Q1 in Ergonomics]** DOI
  Integrated behavioral, eye-tracking, and NASA-TLX evidence to examine interface support in human-AI feedback tasks.
- Du, Y., Zhang, J., & Xue, C. (2025). Research on the Multi-Chart Layout Design of Large-Screen Interface: A Perspective on Cognitive Load. HCII 2025, LNCS 15766, 31-46. DOI
  Evaluated the effects of chart quantity, border style, and layout on cognitive load and visual search using a Unity platform and linear mixed models.

MANUSCRIPT SUBMITTED

- Du, Y., Chen, Y., & Xue, C. (2026). *Designing Corrective Feedback: How Interface Structure Shapes User Review, Training Data, and Model Outputs.* Manuscript submitted to *ACM Transactions on Computer-Human Interaction (TOCHI)*, August 2026.
  Examines how interface structure shapes user review behavior, feedback-data quality, and downstream model outputs through 1,026 traceable records and matched LoRA evaluations.

RESEARCH EXPERIENCE

RLHF Interface Design and Interaction Optimization | Master's Thesis Nov. 2024 – May 2026
- Designed an SMM-supported feedback framework and conducted paired-comparison and function-level studies using behavioral, eye-tracking, and subjective measures.
- **Outcome:** Collected 1,026 valid records (828 strong-signal), built paired feedback datasets, and fine-tuned Qwen2.5-7B with LoRA; evaluation showed improved preference and multidimensional ratings for the model trained on the interface-mediated dataset.

Leadership Cockpit Human-Machine Interface Optimization | Core Team Member Oct. 2023 – Jan. 2025
- Designed interface prototypes and cognitive-load experiments for a leadership cockpit; translated findings into visualization and layout refinements. Related work was accepted at HCII 2025.

Confidential Application Platform Project | Core Team Member Feb. 2024 – Jan. 2026
- Contributed to system architecture, interaction logic, visual design, and workflow mapping; delivered a software interface covering the complete task process.

Static Comfort Evaluation System of Driver Seats | Core Team Member May 2022 – Oct. 2022
- Standardized 3D seat models from point-cloud data using CATIA and delivered modeling documentation for the enterprise digital model library.

PROFESSIONAL EXPERIENCE

Shenzhen Shokz Co., Ltd. | IT Product Management Trainee **Jul. 2026 – Present**
- Translate user-research workflows and stakeholder needs into product requirements and delivery plans; coordinate validation, UAT, and cross-functional launch of an internal research platform.

CISDI Group Co., Ltd. | Modeling and Drafting Technician Jul. 2022 – Aug. 2023
- Produced and reviewed approximately 300 A0 mechanical drawings and parametric models, supporting on-time engineering delivery.

Langjian Yuli Technology Co., Ltd. | Interaction Designer Jun. 2022 – Jul. 2022
- Translated user needs into high-fidelity interface prototypes and coordinated implementation with engineers and clients.

RESEARCH & PRODUCT SKILLS

- Programming & AI: Python, C#, Unity, LoRA fine-tuning, feedback-dataset construction and audit
- Research & Analysis: Controlled experiments, eye tracking, usability testing, mixed-methods research, linear mixed-effects models, SPSS
- Product & Prototyping: Requirements analysis, workflow modeling, product specifications, prototype validation, UAT, cross-functional delivery, Figma, Pixso
- Design Tools: Photoshop, SolidWorks, CATIA
- **Language:** IELTS: Overall 6.5; all components 6.0 or above

HONORS & AWARDS

- Outstanding Graduate of Southeast University
- National Scholarship (Top 1%) and National Encouragement Scholarship (Top 1%)
- Gold Award, 7th China International 'Internet+' Innovation & Entrepreneurship Competition
- Bronze Award, 13th 'Challenge Cup' Entrepreneurial Plan Competition
- Advanced Individual in Innovation Capability Improvement of Chongqing Municipality
- First-Class Academic Scholarship of Chongqing University (Top 5%)

LEADERSHIP & ACTIVITIES

- Leadership: Archives Docent Team lead; Debate Team vice captain; Graduate Union member; grade assistant.